

	Course Name:	Data Communication Networks	Course Code:	EE317
	Program:	Electrical Engineering	Semester:	Spring 2018
	Duration:	3 Hour	Total Marks:	75
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	Section:	All	Page(s):	3
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Instruction/Notes: 1. Write your Name and Roll Numbers in above space.

50, 7, 15

Question No. 1

Marks: 5+10

CLO 01 Recognize the mathematical model of physical layer and its attributes.

- At most how many symbols per second can be successfully transmitted over a channel with bandwidth 10 kHz and signal-to-noise ratio 20 dB?
- Five channels, each with a 100 kHz bandwidth, are to be multiplexed together. What is the minimum bandwidth of the link if there is a need for a guard band of 10 kHz between the channels to prevent interference? Also draw a figure to show the multiplexed channels.

Question No. 2

Marks: 5+10

CLO 02 Examine the ARQ Protocols as to their effectiveness.

- The following Hamming coded (single bit correction) string was received: 010101010111. Is there an error? If yes, in which position?

$2t+1 = d_{min}$; $d_{min} = t+1 \rightarrow$ detection

	0	1	0	1	0	1	0	1	0	1	1	1
Bit position	1	2	3	4	5	6	7	8	9	10	11	12

- Suppose that a sender and a receiver are using ARQ to perform reliable data delivery.
 - In a Go-Back-N sliding window protocol, the window size is 6. Frames with sequence numbers 1, 2, 3, 4 and 5 have been sent. The sender just received an ACK for frame 1. Frames 6, 7, 8, 9 and 10 are waiting to be sent. Draw the time diagram showing this scenario.
 - Which frame(s) can the sender send before it must wait for the next ACK from the receiver? Explain briefly.
 - Sometime later, the sender transmitted frames 20, 21, 22, 23, 24, and 26; however, frame 22 got lost. If Go-Back-N is used, what frame(s) would the sender have to retransmit? What frame(s) would the sender need to retransmit if Selective-Repeat ARQ was used instead? Explain briefly.
 - Frames of 1000 bits are sent over a 1-Mbps channel using a geostationary satellite whose one way propagation time from the earth is 270 msec. Three-bit sequence numbers are used. What is the maximum achievable channel utilization for Stop-and-wait and Selective Repeat?

$2^3 - 1 = 0 - 7$

below represents the single carriage-return character that was contained at that point in the HTTP header). Answer the following questions, indicating where in the HTTP GET message below you find the answer.

```
GET /cs453/index.html HTTP/1.1<cr><lf>Host: gaia.cs.umass.edu<cr><lf>User-Agent: Mozilla/5.0 (Windows;U; Windows NT 5.1; en-US; rv:1.7.2) Gecko/20040804 Netscape/7.2 (ax) <cr><lf>Accept: text/xml, application/xml, application/xhtml+xml, text/html;q=0.9, text/plain;q=0.8, image/png,*/*;q=0.5<cr><lf>Accept-Language: en-us,en;q=0.5<cr><lf>Accept-Encoding: zip,deflate<cr><lf>Accept-Charset: ISO-8859-1,utf-8;q=0.7,*;q=0.7<cr><lf>Keep-Alive: 300<cr><lf>Connection:keep-alive<cr><lf><cr><lf>
```

- i. What is the URL of the document requested by the browser?
- ii. What version of HTTP is the browser running?
- iii. Does the browser request a non-persistent or a persistent connection?
- iv. What is the IP address of the host on which the browser is running?
- v. What type of browser initiates this message? Why is the browser type needed in an HTTP request message?

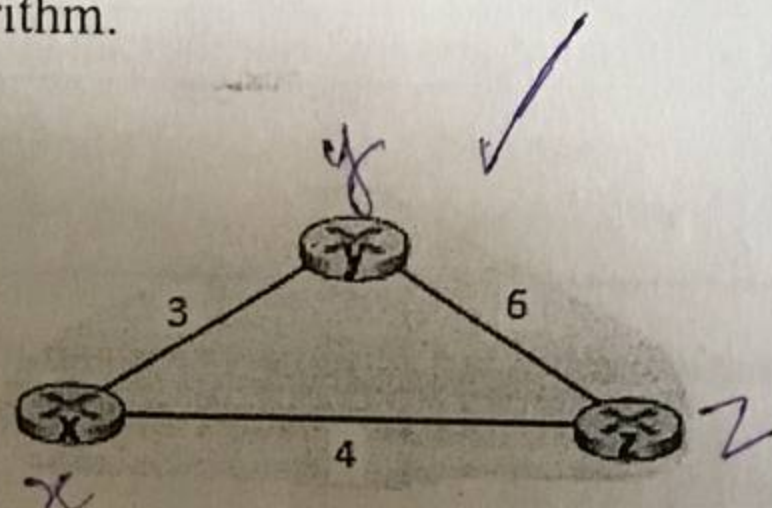
The text below shows the reply sent from the server in response to the HTTP GET message in the question above. Answer the following questions, indicating where in the message below you find the answer.

```
HTTP/1.1 200 OK<cr><lf>Date: Tue, 07 Mar 2008 12:39:45GMT<cr><lf>Server: Apache/2.0.52 (Fedora)<cr><lf>Last-Modified: Sat, 10 Dec 2005 18:27:46<cr><lf>ETag: "526c3-f22-a88a4c80"<cr><lf>Accept-Ranges: bytes<cr><lf>Content-Length: 3874<cr><lf>Keep-Alive: timeout=max=100<cr><lf>Connection: Keep-Alive<cr><lf>Content-Type: text/html; charset=ISO-8859-1<cr><lf><cr><lf><doctype html public "-//w3c//dtd html 4.0 transitional//en"><lf><html><lf><head><lf><meta http-equiv="Content-Type" content="text/html; charset=iso-8859-1"><lf><meta name="GENERATOR" content="Mozilla/4.79 [en] (Windows NT 5.0; U) Netscape]"><lf><title>CMPSCI 453 / 591 / NTU-ST550A Spring 2005 homepage</title><lf></head><lf><much more document text following here (not shown)>
```

- i. Was the server able to successfully find the document or not?
- ii. What time was the document reply provided?
- iii. How many bytes are there in the document being returned?
- iv. What are the first 5 bytes of the document being returned?
- v. Did the server agree to a persistent connection?

CLO 05 Evaluate network layer protocols and their impact on network performance.

- a) A large number of consecutive IP addresses are available starting at 204.18.0.0. Three organizations A, B and C request 2048, 1024 and 4096 addresses respectively in the given order. For each of these, what is the first IP assigned, last IP assigned and the mask in w.x.y.z/s notation?
- b) Consider the three-node topology shown in Figure below. The link costs are $c(x,y) = 3$, $c(y,z) = 6$, $c(x,z) = 4$. Compute the distance tables after the initialization step and after each iteration of a synchronous version of the distance-vector algorithm.



CLO 06 Recognize the model of transport layer and its performance metrics.

- a) Suppose that the two measured SampleRTT values are 108 ms and 124 ms. Assume that the value of EstimatedRTT was 102 ms and DevRTT was 5 ms just before the first of these two samples were obtained. Using the typical values of $\alpha = 0.125$ and $\beta = 0.25$, compute the EstimatedRTT and DevRTT after each sample is obtained. Also compute the TCP TimeoutInterval after each of these samples is obtained.

- b) Data is being transferred between Host A and Host B over a TCP session. Host A and Host B have allocated TCP Receive Buffer spaces of 5 kbytes and 10 kbytes respectively for this session. Data till byte number 1008 has been received at Host B and the byte that has been read by the application layer at Host B is 90. Host A has sent data up to byte number 2043 while the byte that has been acknowledged by Host B is 1999. Data till byte number 2108 has been received at Host A and the byte that has been read by the application layer at Host A is 180. Host B has sent data up to byte number 1999 while the byte that has been acknowledged by Host A is 1960. Ignoring TCP congestion control, compute TCP receive window and TCP sender window at both Host A and Host B.

CLO 07 Research existing network applications and future trends.

- a) Consider an HTTP client that wants to retrieve a Web document at a given URL: <http://www.nts.org.pk>. The IP address of the HTTP server is initially unknown. Plot a hierarchy of DNS name servers and show retrieval of the IP address of this website by the client if iterated queries are employed.
- b) Consider the following string of ASCII characters that were captured by Wireshark when the browser sent an HTTP GET message (i.e., this is the actual content of an HTTP GET message). The characters `<cr>` `<lf>` are carriage return and line-feed characters (that is, the italicized character string `<cr>` in the text